

Exploring the Interpretability of Deep Learning Models for Breast Cancer Diagnosis using XAI Techniques: A Comparative Study with Expert Radiologist Interpretations



MS Thesis

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**COMSATS University Islamabad
Pakistan
Spring 2023**



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A thesis submitted to
COMSATS University Islamabad

In partial fulfillment
of the requirement for the degree of

Master of Science
in
Health Informatics

by

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DEDICATION

I dedicate this thesis to all the beings whose lives have been affected by breast cancer. To the patients who have battled heroically, the fighters who have shown resilience, and the families and friends who have provided unwavering support throughout the journey.

ACKNOWLEDGEMENTS

Primarily, I would like to thank Allah Almighty for providing me with this wonderful opportunity to pursue and complete my MS research. I would like to express my sincere gratitude to my supervisor Prof. Dr. Haroon Ahmed whose unwavering trust in my abilities and insightful guidance enabled me to reach this point. Collaborating with him has been a great privilege for me.

I would also like to thank all my teachers at the Department of Health Informatics of COMSATS University, Islamabad, Pakistan, who built my theoretical and conceptual knowledge. I would also like to thank my research group colleagues, especially Fawad Ahmad whose expertise has shaped the direction of this research and has been invaluable in this research.

Finally, I would like to express my deepest gratitude to my family and friends for their efforts, encouragement, motivation, prayers, and continuous support throughout my journey. Their love and belief in me have been instrumental in achieving this milestone.

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ABSTRACT

Exploring the Interpretability of Deep Learning Models for Breast Cancer Diagnosis using XAI Techniques: A Comparative Study with Expert Radiologist Interpretations.

BY

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Breast cancer is a serious medical condition that requires early identification for effective treatment. Mammography is a widely used imaging technique for breast cancer screening, but its analysis can take time and be subjective. Deep learning algorithms have shown remarkable contributions in this area. However, their lack of interpretability creates a noteworthy challenge for their incorporation into medical practice. This study aims to explore the interpretation of deep learning models for breast cancer detection using Explainable Artificial Intelligence (XAI) technology and conduct a comparative study with expert radiologist Interpretations.

The research employs XAI approaches, namely the Grad-CAM (Gradient-weighted Class Activation Mapping) algorithm, to visualize and comprehend the characteristics and areas of concern to the results of deep learning networks. The models' interior workings are made clearer and more understandable. by creating heat maps that emphasize the relevant spots in mammograms.

The comparative study entails evaluating the diagnostic accuracy of machine learning systems and comparing it to experienced radiologists' analyses of mammography datasets. Performance indicators such as accuracy, sensitivity, specificity, and agreement measurements will be employed for evaluation. This comparison tries to find areas of agreement and disagreement between deep learning models and expert radiologists, revealing the strengths and limits of each technique.

Furthermore, the study seeks to discover the elements that influence deep learning model predictions in breast cancer detection. Analyzing the mammography features, patterns, and characteristics that contribute significantly to the model's predictions may offer insight into the crucial picture aspects examined by the models, leading to the discovery of novel diagnostic markers for breast cancer.

This study's outcomes will help to improve the interpretability of deep learning models, increase trust and confidence in AI-driven decision-making, and discover prospects for improving breast cancer detection by combining AI and human knowledge. The findings will give suggestions and guidelines for using deep learning models in clinical practice, paving the road for their effective use in real-world healthcare settings.

Table of Contents

Chapter 1 Introduction	1
1.1 Background	1
1.2 Deep Learning Models and Explainable Artificial Intelligence Algorithm ...	6
1.2.1 VGGNet	8
1.2.2 ResNet	9
1.2.3 Grad-CAM	9
1.2 Significance	10
1.2.1 Practical Significance	11
1.2.1 Scientific Significance	11
1.3 Motivation	11
1.4 Problem Statement	12
1.5 Research Objectives	13
1.6 Research Questions	13
1.7 Research Contribution	13
1.8 Thesis Organization	14
Chapter 2 Literature Review	15
Chapter 3 Material and Methods	22
3.1 Dataset	22
3.2 Training and Testing	26
3.3 Methods Used	26
3.4 Preprocessing	27
3.5 Segmentation	28
3.6 Feature Extraction	28

3.7	Data Augmentation	28
3.8	Classification	29
3.9	Evaluation matrix	30
3.10	Explainable Artificial Intelligence survey	35
Chapter 4 Results and Discussion		39
4.1	Results and Discussion	39
4.2	Requirements for Explainable Artificial Intelligence System development.	42
4.2.1	Obligations and Conduct	42
4.2.2	Security and Secrecy	42
4.2.3	Transparency and Skepticisms	42
4.3	Pros and Cons of Explainable Artificial Intelligence	43
Chapter 5 Conclusion		46
5.1	Conclusion	46
5.2	Recommendations	46
References		48

List of Figures

Figure 1.1 Breast Anatomy	1
Figure 1.2 Benign and malignant melanoma	2
Figure 1.3 Mammogram from MIAS Database	4
Figure 1.4 CNN Architecture	7
Figure 1.5 VGGNET Model Architecture	8
Figure 1.6 ResNet Architecture	9
Figure 3.1 Pie Chart description of type of tissues	23
Figure 3.2 Bar chart for severity of cases	25
Figure 3.3 Sample images from the mini MIAS database.	26
Figure 3.4 Flow diagram of methodology.	27
Figure 3.5 Augmented Images of Mais dataset	29
Figure 3.6 Accuracy plot of ResNet50	33
Figure 3.7 Accuracy plot of VGGNet 16	33
Figure 3.8 Heat generated by GradCam.	34

List of Tables

Table 2.1 Review Matrix for Literature Review.....	19
Table 3.1 Character of background tissue.....	22
Table 3.2 Class of abnormality present.....	24
Table 3.3 Data Augmentation.....	29
Table 3.4 Evaluation result of CNN model.....	31
Table 3.5 Evaluation results of ResNET50.....	31
Table 3.6 Evaluation result of VGGNET 16.....	32
Table 3.7 Statistics of the professional survey's rating using the CHF scale.....	35
Table 3.8 result of XAI satisfaction Scale.....	36
Table 3.9 Evaluation Summary of Expert Survey.....	36
Table 4.1 Normal Distribution and reliability scores.....	38
Table 4.2 Regression Model Summary.....	39
Table 4.3 Difference table.....	39
Table 4.4 Pros and Cons of XAI.....	42

List of Abbreviations

XAI	Explainable Artificial Intelligence
GradCAM	Gradient Weighted Class Activation Mapping
DNNs	Deep Neural Networks
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
ANN	Artificial Neural Network
GPU	Graphics Processing Units
CNNs	Convolutional Neural Network
VGGNet	Visual Geometric Group network
ResNet	Residual Network
ILSVRC	Large Scale Visual Recognition Challenge
VQA	Visual Question Answering
ML	Machine Learning
DL	Deep Learning
DLAEABA	Deep Learning Aided Efficient Adaboost Algorithm
DDSM	Digital Database for Screening Mammography
CAD	Computer Aided Diagnosis
ROI	Region Of Interest
DSC	Dice Similarity Coefficient
DCT	Discreate Cosine Transform
PGM	Portable Gray Map
MIAS	Mammographic Images Analysis Society
MCC	Mathews Correlation Coefficients
CHF	Cahour Forzy Scale

Chapter 1

Introduction

1.1 Background

Breast tumors are the most often diagnosed cancer in the world, and their incidence is rising daily. It is the leading cause of death among women. In 2020, 16% of women died from this condition [1]. Pakistan has the highest frequency and fatality rate of breast cancer in Asia, with many patients diagnosed early in life [2]. The deaths from breast cancer are higher in underdeveloped countries. Principal factors that are in the increasing risk of Breast cancer are genetic, environmental, and lifestyle. To decrease the risk of breast cancer lactation and exercise plays important roles [3]. In Pakistan, the fatality rate is one-third yearly. As a result, it is critical to detect and treat cancer in its preliminary stages, lest it extends to the chest or other regions of one's body. Appropriate initial treatment, based on a precise and effective diagnosis, can raise the recovery chance of tumors in the Patients by as much as 80 percent.

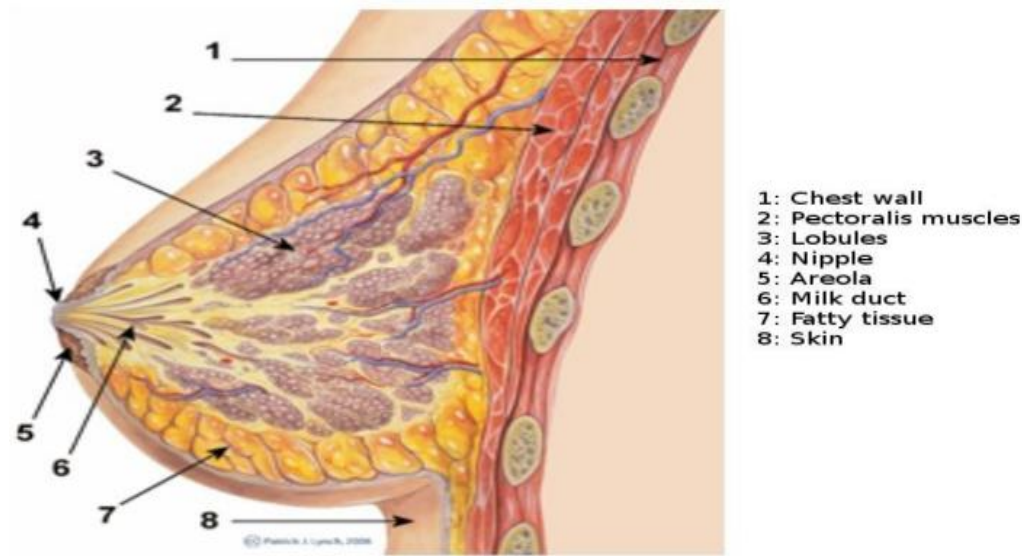


Figure 1.1: Breast Anatomy [4].

The fatty tissue and supporting tissue are mixed to form breasts. A breast's anatomy includes 15 to 20 lobes, each of which is divided into several smaller pieces termed lobules. The tiny bulbs known to produce milk are the lobules' ends. The fluid that contains milk travels through ducts that link the bulb, lobules, and lobes to the nipple. Additionally, the breast is home to neural pathways and bloodstream and lymphatic veins. The mammary gland is supported and kept in form by its connective tissue and ligaments. The Breast anatomy is shown in Figure 1.1.

Melanoma lesions in the breast can be classified as benign or malignant. Both groups have the potential to be malignant or not. For example, benign lesions cause abnormalities in the epithelial cells but cannot develop further. Therefore, they cannot cause breast cancer. Now, appropriately analyzing and classifying benign and malignant tumors in microscopical pictures is difficult. [5].

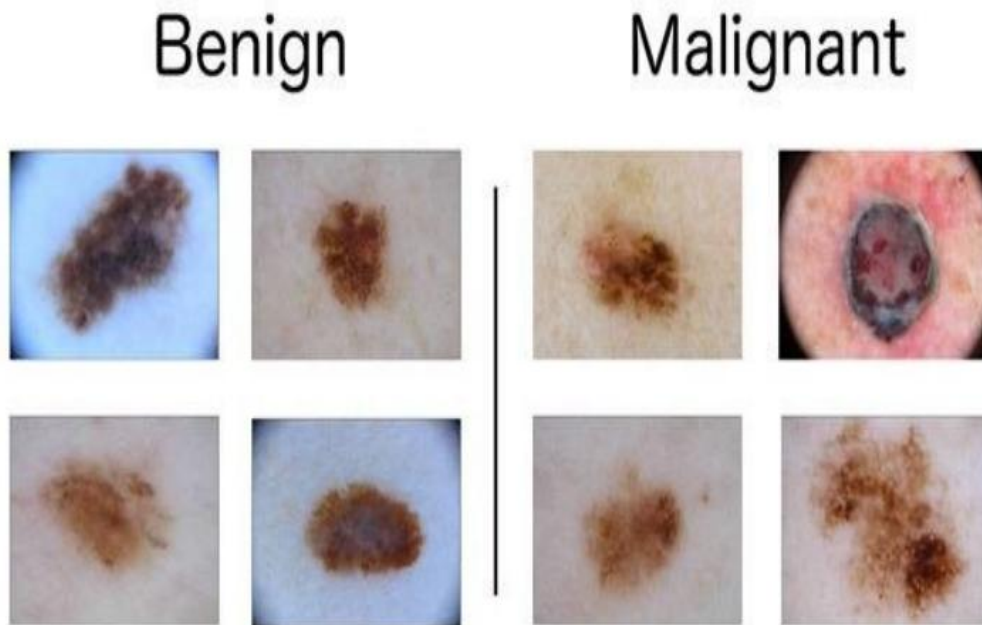


Figure 1.2: Benign and malignant melanoma [6].

Breast cancer can be detected early using a variety of methods, such as screening tests, feeling unwell inside, or going to your local doctor. These tactics slow down the mortality rate and increase the chance of successful therapy [7]. In practice, radiologists use a variety of techniques, including mammography and biopsy, to detect breast cancer [5].

Mammography is an electromagnetic screening technique used to inspect the breasts to check for signs and symptoms of breast illnesses like tumors. It functions like an examination and diagnosing tool. In Screening mammography, every breast is examined twice during the procedure, which is strongly advised for women 40 years of age and older. There should be consideration given to further mammograms if questionable areas seen on the mammography.

Screening mammography is conducted to check for any abnormalities in the breast tissue and to provide the groundwork for further research. In contrast to screening, diagnostic mammography is used when a precise diagnosis is required. In this situation, a more concentrated study is conducted if it finds any indication of a lump or tiny tumor. When a particular breast area is being monitored periodically, diagnostic mammography will also be conducted. Diagnostic mammography can give a wider perspective into the observed abnormalities during the screening phase and provide more information based on the obtained extra pictures that also comprise spot compression or spot compression with magnification.

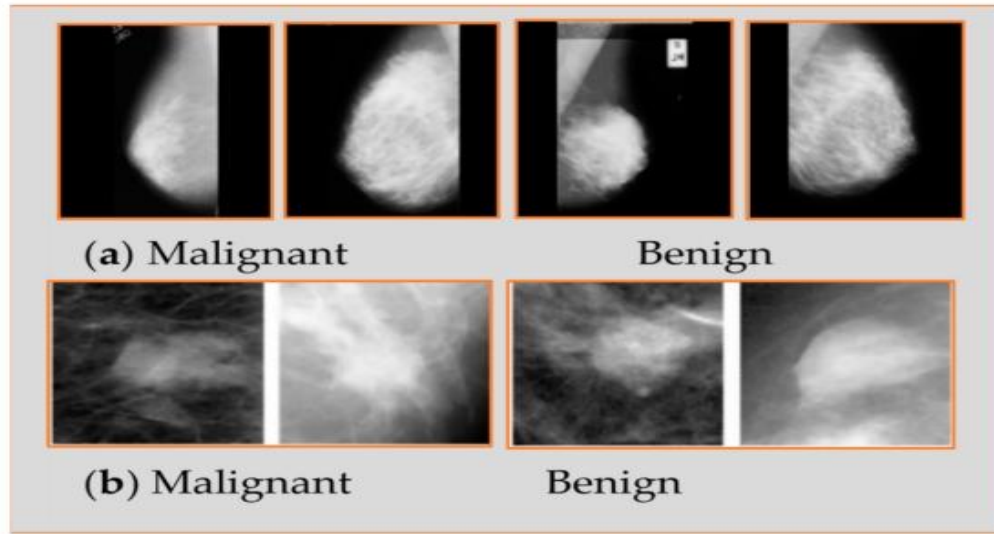


Figure 1.3: Mammogram from MIAS Database [8].

But still, it is exceedingly difficult for highly qualified physicians to give error-free diagnoses in a fleeting time. For decades researchers have been working to find the procedures that will help the radiologist in the quick identification of diseases. Although they are successful somehow there still needs to develop a system that can be used in performing their daily operations. To encounter this problem, Machine learning techniques especially XAI focused to develop systems that may help to improve the efficiency in medical imaging classification.

The fast development of deep neural networks, and multi-layer neural network algorithms, exemplifies the exceptional performance of artificial intelligence (DNNs). Nonetheless, many who are interested in and employing artificial intelligence are still baffled by the black box problem that is arising with DNNs. Artificial intelligence must be more comprehensible, explicable, and user-oriented for critical medical imaging systems to provide the best potential human-machine interaction. The capacity to understand the context and extrapolate the meaning of a topic or concept from a limited dataset is a distinguishing characteristic of artificial intelligence. But, making sense of a vast volume of rapidly advancing, unstructured data is challenging.

Thankfully, artificial learning techniques and other technologies enable artificial intelligence systems to draw conclusions, make recommendations, or take other actions from a wealth of data. Nevertheless, these tools have several issues, including the necessity for big, high-quality, and widely available datasets with labels, the demand for a significant amount of technical work [9], the non-linear quality of DNNs, and the challenge of connecting data to algorithms. While many researchers want to employ artificial intelligence to enhance decision-making and enable autonomous movement, (XAI) provides a technical explanation of explainability for key participants including creator, utilizer, administrators and those making the decisions. XAI studies and creates internal reasoning procedures for machine learning platforms aims to improve models based on artificial intelligence while making algorithmic results easier to comprehend beneficiaries. Systems for medical imaging are crucial for the precise identification and diagnosis of illnesses. Doctors or radiologists must reach an ongoing, all-pervasive condition of trust cycle. Regarding advances in diagnostic scanning innovation, radiologists may examine their job procedures from a variety of perspectives, support interaction built on commitment efficiently, and differentiate among proactive trust and process evaluation. Diagnostic methods other than clinical imaging, such as computed tomography (CT) and magnetic resonance imaging (MRI), are being compared.

Compared to other photos, gig pixel images are more complicated. It is known that the characteristics of large-scale histology pictures differ significantly from those of ImageNet images and radiological images. This work intends to give much-needed proof and clarify future research paths of the use of deep learning algorithms on histological pictures, as well as to give radiologists information that is understandable. Human radiologists regularly analyze and classify a sizable number of slides from diverse sources. This is a difficult and occasionally impossible task. The difficulty in detecting illnesses in humans with highly distinct physiological and anatomical features might be partially related to the fact that few professionals have ability in doing so. Lately, the application of artificial intelligence in pathology has increased the diagnosis speed and accuracy of both human and veterinary radiologists. Expert radiologists must conduct a thorough and objective analysis to properly understand the claims produced by the

algorithm to determine the effectiveness and reliability of XAI. In certain instances, the evaluation's effectiveness is constrained by time constraints or the radiologists' inadequate ability. With the advancement of artificial intelligence systems, this dilemma may be resolved.

1.2 Deep Learning Models and Explainable Artificial Intelligence Algorithm

Deep learning is a sort of artificial neural network (ANN) that match a mufti-layered human cognitive system. It is a subset of machine learning. Deep learning is gaining prominence as it is used to massive volumes of health care data. Even though ANN was invented in 1950, its training met considerable constraints and difficulties for problem solving due to concerns such as a lack of computer capability and adequate data. However, with the current availability of vast data, greater computing power with graphics processing units (GPU), and new methodologies for training a deep neural network (DNN), many constraints and problems have been addressed. Deep learning algorithms outperform humans in a variety of fields, including medical imaging, in this context.

Image and video processing are widely used in medical science. Biomedical imaging methods are crucial in both diagnostic and therapy. These strategies have aided in the improvement of patients' health care. With greater precision in illness identification and surgical operations, image-guided therapy has lowered the chance of human mistake. The history of clinical imaging begun in the mid-1890s and has steadily progressed owing to today's technology backbone and readily available statistics [10].

Medicine is an essential practical field, as well as the most challenging endeavor for AI and deep learning. One of the difficulties in using deep learning and artificial intelligence in the field of medicine is dealing with multidimensional picture databases that are insufficient, problematic, wrong, and absurd [11]. The challenge's many facts about XAI are considered in the context of a broader view of medicine as the ultimate strategy.

Especially if it comes to clinical images, doctors must understand the rationale behind how an artificially intelligent system gets a decision.

Convolutional neural networks (CNNs), initially described by Fukushima have gained prominence in recent years as a particularly promising kind of deep learning. Convolutional layers, which have their roots in image processing, have effectively entered all subfields of deep learning. Figure 1.4 shows the standard components of a CNN, which include a neural layer(s), pooling or samples layer(s), stimulation function(s), connections, and regulating layer(s). Convolutional operations' primary goal is to acquire mappings of features using the source images. Compared with many other algorithms, CNNs are more often employed, notably in image-processing systems. VGGNet and ResNet, the two popular CNNs, are employed in this work [12].

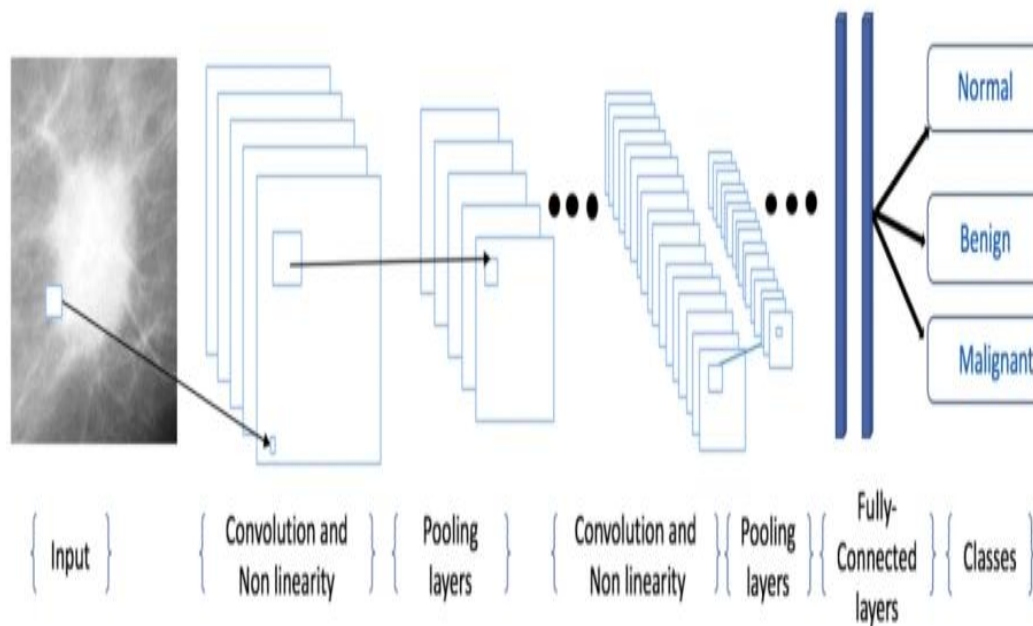


Figure 1.4 CNN Architecture (Deep convolutional neural networks for mammography: advances, challenges, and applications).

1.2.1 VGGNet

VGGNet, suggested by Simonyan and Zisserman, finished second at the 2014 ILSVRC competition. This network's high performance is due to a fundamental aspect: the mesh's depth. The network is streamlined across the VGGNet by keeping only the technique of convolution with a 3x3 window and a 22 filter without modifying the depth. The last fully connected layer of VGGNet comprises 1,000 nodes, as illustrated in Figure 1.5, and its classification (output) layer includes a soft-connected (SoftMax) layer [13].

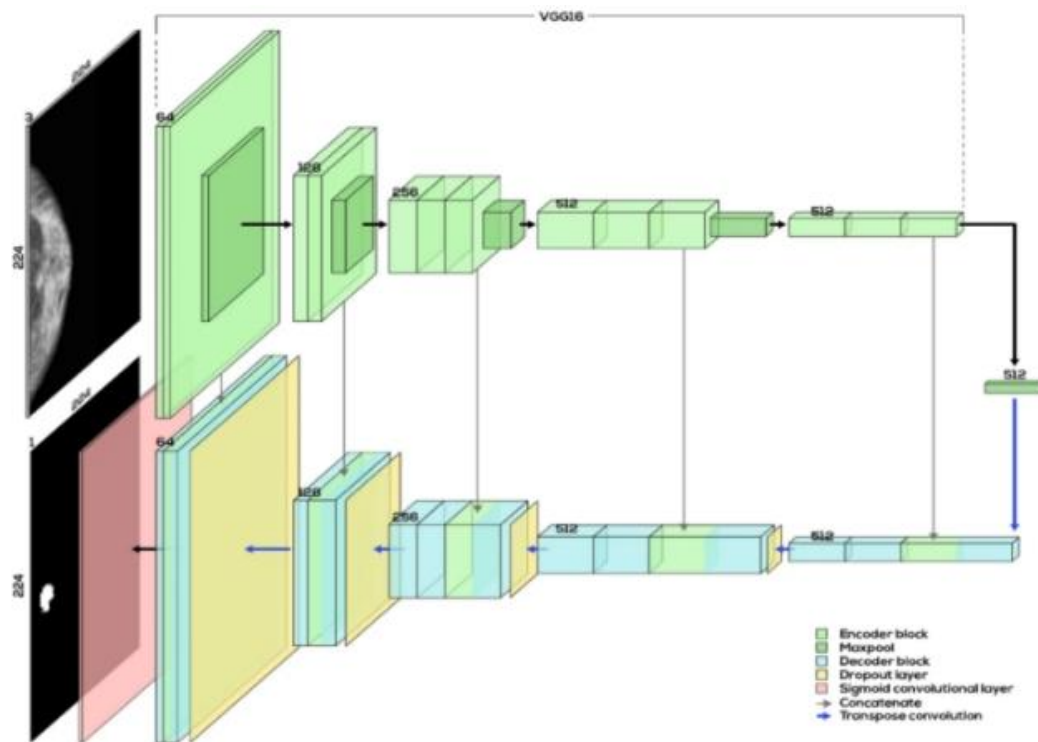


Figure 1.5: VGGNET Model Architecture

(https://miro.medium.com/v2/resize:fit:2000/format:webp/1*VmIjedEapD13yznguXcqTw)

1.2.2 ResNet

The CNN ResNet won the 2015 Large Scale Visual Recognition Challenge (ILSVRC). This design heavily relies on specific jump connections and batch normalization to minimize parameters and enhance depth, which dramatically improves training success. ResNet features distinct residual modules, as seen in Figure 1.6 [14].

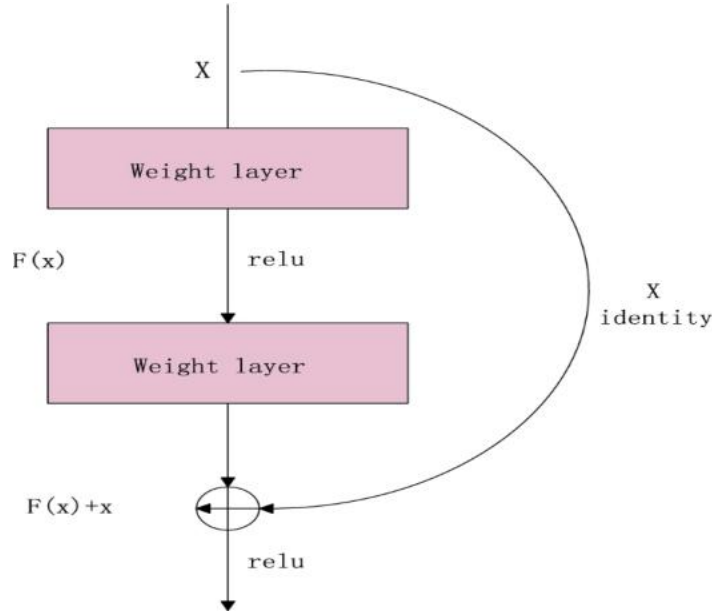


Figure 1.6: ResNet Architecture (<https://link.springer.com/bookseries/7818>)

1.2.3 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) employs gradients from any target idea to generate a crude location map highlighting important regions in the image for identification. Grad-CAM can be applied to a wide variety of CNN model families. Grad-CAM is used in conjunction with extremely fine representation to supply an excellent quality class-discriminate display, which is then utilized to train readily available image classification, subtitles, and visual question answering (VQA) models. The tool grows as a generalization of class activation. The primary distinction between

Grad-CAM and CAM is the use of global average pooling (GAP) and soft-max activation following convolution layers [15]. Grad-CAM offers considerable localization for all image classes. Localization in an input picture is generated by following the gradients in CNN back propagation. Heat maps are produced on the most significant characteristics characterizing a class in this manner.

1.2 Significance

The lack of interpretability of the models is a critical issue as AI-based detection tools for cancer of the breast gain popularity. Deep learning models, for instance, are well known to be fully correct, but because of the opaqueness of their internal workings, it might be difficult for doctors to comprehend how the models arrived at their diagnosis. The safety, dependability, and morality of using AI-based diagnostic tools in clinical contexts are questioned by this lack of interpretability.

This project's purpose may be to investigate the interpretability of deep learning models for the detection of breast cancer utilizing XAI approaches. By doing this, you may aid in the creation of more understandable and transparent AI-based diagnostic tools, which can result in improved patient outcomes along with a more accurate diagnosis.

A large set of characteristics could enhance XAI classifier classification precision. These derived characteristics, on the other hand, can be useful for picture comparison or identification methods, as well as training in guided systems. The application of the XAI approach decreases false positives and enhances diagnosis rates in the initial stages of breast cancer. A highly correct classification machine learning system may assist in timely identification of malignancies in the breast and offer additional perspectives for the radiologists and other medical specialists, decreasing female mortality and minimizing the number of unneeded breast biopsy. Furthermore, prompt medical care can be supplied to save lives.

1.2.1 Practical Significance

The importance of this study exists in its potential to increase both the transparency and precision of AI-based breast cancer detection systems. Although deep learning models have proven encouraging outcomes in the detection of breast cancer, both patients and physicians are concerned about their lack of interpretability. This problem may be resolved and the confidence and adoption of AI-based diagnostic tools in clinical settings can be raised by using XAI approaches to evaluate deep learning models.

1.2.1 Scientific Significance

This study can also advance our understanding of the value of interpretability in AI-based diagnostic tools and the expanding body of information on the application of AI in health care. This study can aid in directing the development of deep learning systems by showing the advantages of utilizing XAI approaches to enhance the interpretability of deep learning models.

1.3 Motivation

The advancements achieved in the field of artificial intelligence (AI) during the earlier decade have resulted in a remarkable rise in the reliability of most applications for machine learning. Medical image analysis is one of the fields where advances in technology have ensured human-level accuracy in the classification of various forms of health data. Breast cancer screening tools currently available have high false positive rates, as well as low accuracy, sensitivity, and specificity [16]. This results in an enormous influence on the female estimation mortality rate.

The primary benefit is in addressing the problem of detecting distinct types of breast lumps. Because of the minimal contrast, the existence of pectoral muscle within the breast region, and the comparable size range of pectoral muscle, breast tissues, and masses, microcalcification and lumps appear challenging to recognize [17]. By investigating the interpretability of deep learning models, researchers can increase the diagnostic accuracy and reliability of automated systems for breast cancer diagnosis,

resulting in better patient care and treatment. Expert radiologists are critical in the diagnosis of breast cancer, and their interpretations are regarded the gold standard.

This research intends to analyze the performance of these models in comparison to human experts, find areas of agreement or difference, and highlight the complementing benefits of each technique by performing a comparative analysis between deep learning models and expert radiologists. This comparative study can supply light on the possible function of AI systems in assisting radiologists and boosting diagnosis accuracy. Overall, the motivation for pursuing this project stems from the possibility of improving breast cancer diagnosis, improving the interpretability of deep learning models, conducting a comparative analysis with expert radiologists, advancing explainable AI in medical imaging, and contributing to scientific understanding, all of which could lead to improved patient experiences and healthcare practices.

1.4 Problem Statement

Deep learning models for breast tumor diagnostics have shown encouraging outcomes in terms of accuracy and efficiency. Yet a deficiency of interpretability and vulnerability in these models makes their adoption and integration into clinical practice difficult. Deep learning models' black box nature makes it difficult to grasp how they arrive at their predictions, decreasing the trust and belief of medical personnel and their patients.

Furthermore, while deep learning models have shown great accuracy in malignancy detection, it is critical to compare their performance to that of experienced radiologists, who are regarded as the gold standard in this field. Discovering where deep learning models and professional radiologists agree and disagree might give a better understanding of the potential upsides and drawbacks of AI systems in helping radiologists and enhancing diagnosis accuracy [18].

As a result, the topic addressed in this work is the necessity to investigate the interpretability of deep learning models for breast cancer detection utilizing XAI approaches, as well as perform a comparative study with expert radiologist

interpretations. The aim is to look at the inner workings of deep learning models, identify the elements that influence their predictions, and compare their performance against experienced radiologists. This project seeks to improve the interpretability of deep learning models, boost trust in AI-driven decision-making, and uncover the potential to improve breast cancer detection through the collaboration of AI and human knowledge [19].

1.5 Research Objectives

The study is aimed at supplying much-needed data and clarifying future research paths regarding the application of machine learning algorithms on Mammograms. The research objectives are:

- To evaluate the interpretation of deep learning model for breast cancer diagnosis using XAI techniques.
- To supply insight into how well these models can be understood and interpreted by medical professionals.

1.6 Research Questions

1. How could XAI methods like Grad-CAM be used to explain deep learning models' decision-making processes for breast cancer diagnosis?
2. How do XAI-generated explanations compare to professional radiologist interpretations of the same mammography images?
3. What are the advantages and disadvantages of utilizing XAI approaches to improve interpretability in deep learning algorithms for breast cancer detection?

1.7 Research Contribution

By studying the interpretability of deep learning models using XAI approaches and comparing the offered explanations to expert radiologist interpretations, this work will

aid to progress the field of breast cancer diagnosis. The study will primarily give insights into the efficacy and comparability of XAI methodologies for explaining deep learning model decision-making processes in the context of a breast cancer diagnosis. The study will contribute to enhanced trust and acceptance of deep learning models in clinical practices by assessing in their interpretability and comparing it to expert radiologist interpretations, which can lead to increased accuracy and efficiency in breast cancer diagnosis.

The research will also contribute to the body of knowledge on the use of AI technology in healthcare, as well as the importance of interpretability and transparency in AI decision-making. The study's findings can be used to develop future XAI methods and approaches for improving the interpretability and explanation of deep learning models in health care and beyond.

1.8 Thesis Organization

The remaining of the thesis is organized as follows:

Chapter 2 has literature review. Chapter 3 includes the research material and methods. Chapter 4 describes the results of our proposed method and discussion. Finally, the conclusion with future work is presented in Chapter 5.

Chapter 2

Literature Review

Numerous studies were conducted around the globe considering various livestock owners and acquiring knowledge regarding their issues, knowledge, opinions, and behavior regarding lumpy skin disease, its screening, and control.

A thorough study on the application of ML and DL algorithms to identify breast cancer was provided by Chugh et al. They reviewed the literature review and studies linked to the categorization of breast cancer in detail. They also emphasized both the advantages and disadvantages of the features of these strategies. Incorporating the results of other research, the researchers involved in this study concluded that Deep Learning- based methods are much better for categorizing breast cancer photos only for larger data sets [20].

To categorize breast cancer photos into benign and malignant tumors, Khan et al. used the ideas of transfer learning and deep learning. Using pre-trained ResNet, GoogLeNet, and VGGNet architectures, they determined the features. By applying the average pooling technique, these features were also made available to the fully linked layers. For their study, they used a dataset of 8000 photos in total; out of which 6000 were used to train the network and the remaining 2000 were used for testing. Their proposed strategy had a maximum average accuracy of 97.25 percent [21]. In their study, Khan et al. suggested utilizing DCNN to categorize photos of charred human skin. Their dataset included 450 photos. They used 65% of the photos to train the network and 35% of the photos to evaluate its accuracy. As a result, they classified the photos with 79.4% accuracy [22]. In another study, the dataset, which originally comprised 600 photos, was expanded. The segmentation and categorization of charred human skin was done using the same method. The accuracy of the classifier was trained and assessed with a ratio of 60% and 40%. As a result, they classified the photos with an accuracy of 83.4 % [23]. Using various supervised machine learning techniques, Gupta et al. sorted breast cancer

photos into malignant and non-malignant tumors. Their algorithms included SVM, k-nearest neighbor, random forest, decision tree and logistic regression. The accuracy was 98.24%, also using the Adam Gradient Descent Learning models [24].

In-depth research on the classification of breast cancer pictures using deep learning and imaging methods was presented by Murtaza et al. They extracted 49 distinct research papers from 8 diverse sources that had been presented at conferences and publications. They emphasized every ML and DL algorithm used to various medical picture classification tasks, such as the classification of breast cancer. For new researchers, they offered the 10 open research challenges, which included the automated identification of breast cancer [25].

Using bidirectional encoder representations (BER), Zhang et al. presented the clinical data of breast cancer patients. They have also extracted characteristics or features from the infected photos of breast cancer patients using deep learning algorithms. The greatest results obtained with their claimed methods were 96.73% [26]. Machine learning-based picture segmentation and classification has been the subject of several research. In their research for the detection and classification of breast cancer, Zheng et al. introduced a mathematically realized Deep Learning aided Efficient Adaboost Algorithm (DLAEABA) with the capability of advanced level calculations. Also, they employed a few computer vision methods. Also, they provided a comparison by demonstrating the experimental results' 97.2% accuracy [27].

A deep learning detection and classification integrated CAD system is suggested with the goal of improving the diagnosis performance of breast lesions from all digital X-ray mammograms. The YOLO detector detects breast lesions with an F1-score of 99.28% for DDSM and 98.02% for INbreast. The YOLO detector enhanced the modified InceptionResNet-V2 classifier, resulting in promising diagnostic performance with overall accuracies of 97.50% for DDSM and 95.32% for INbreast. In less than 0.025 seconds, the suggested deep learning CAD system can detect and categorize breast abnormalities in a single mammogram [28].

The work shows that deep learning models trained end-to-end may be extremely accurate and potentially easily transferrable across various mammography systems. Deep learning approaches have immense potential to increase the reliability of breast cancer diagnosis on screening mammography as training datasets and computational resources become more available. Our method might aid in the future development of superior CAD systems that could be used to help prioritize the most worrisome cases to be reviewed by a radiologist, or as an automated second reader after an initial independent interpretation. Our end-to-end technique may also be used to other medical imaging situations with limited ROI markings [29].

Human domain knowledge is difficult to include for breast tumor diagnosis because form, border, curves, magnitude, and other typical clinical assumptions differ widely between individuals and cannot be used. The study describes a novel approach to incorporate visual prominence into a deep learning algorithm for segmenting breast tumors in ultrasound images. Image maps with visual saliency feature places which are more probable to capture radiologists' interest. The suggested technique includes emphasis segments into a U-Net design and develops visualizations of features that prioritize geographic locations of high saliency. The validation results demonstrate that algorithms with salient attention layers perform better at tumor segmentation. On a batch of 510 images, the approach produced a Dice similarity coefficient (DSC) of 90.5% [30].

Traditionally diagnosing brain tumors is an error-prone and time-consuming operation for radiologists; consequently, an automated method is essential. Although the binary classification approach, such as malignant or benign, is extremely straightforward, radiologists struggle to identify multifunctional tumors (T1, T2, T1CE, and Flair). We describe an automated multimodal classification system for brain tumor type categorization based on deep learning. The suggested approach is made up of five main phases. Linear contrast broadening is performed in the initial phase using edge-based histogram equalization and the discrete cosine transform (DCT). In the next stage, deep learning extraction of features is performed. For feature extraction using transferred

learning, two pre-trained convolutional neural network (CNN) models, VGG16 and VGG19, were used [31].

This study developed an innovative blended strategy for Tuberculosis X-ray picture categorization. The technique entails extracting features from chest X-ray images with a MobileNet deep neural network and then sorting the outcome's enormous number of features with the recently proposed Artificial Ecosystem-based Optimization (AEO) method to include the most notable features and eliminate features that are unnecessary. The categorization approach (MobileNet-AEO) was validated using two publicly available records, Shenzhen Data and Dataset 2, each of which contain chest X-ray images. With adequate categorization precision, MobileNet-AEO functioned brilliantly. In addition, the complexity was lowered, which improved computing time. MobileNet-AEO was able to reduce the number of features from 50 K to 25 and 19 while maintaining an accuracy of 90.2% and 94.1% [32].

Table 2.1: Review Matrix for Literature Review

No.	Authors	Description of Research	Methodology	Dataset	Evaluation Measures	Limitation	Year
1	Chugh, G., Kumar, S. & Singh, N.	This study examines several computational modeling and advanced algorithms classifiers for identifying breast cancer.	computer-aided diagnostic (CAD) systems that use deep learning (DL) and machine learning.	Image data set is used from open source	examines several classifiers from both deep learning and machine learning methods for detecting breast cancer	Conventional study, no new system proposed.	2021
2	SanaUllah Khan, Naveed Islam, Zahoor Jan, Ikram Ud Din, Joel J. P. C Rodrigues	For the categorization of tumors in chest histopathology pictures, a deep learning-based system is suggested. It provides a high degree of accuracy.	The CNN architecture was utilized to identify and categorize tumor cells in breast cytology pictures.	Breast cytology images taken as data set.	GoogLeNet, VGGNet, and ResNet, three distinct deep learning architectures have each been examined.	Normal and malignant cells have erratic appearances; manually analyzing microscopic pictures is an incredibly difficult and complex process.	2019
3	Puja Gupta, Shruti Garg,	Using their hyper-parameters for the Wisconsin Breast Cancer Dataset, six machine-learning learning models were evaluated.	The project has made use of the scikit-learn library's pandas, matplotlib, tensor flow, and keras tools.	Datasets retrieved from the Wisconsin Breast Cancer Dataset	Binary cross entropy is used to analyze the system and forecast the true value.	Image based data is not used in it.	2020

4	Murtaza, G., Shuib, L., Abdul Wahab, A.W. et al.	In five areas, including the usage of BrC imaging systems, datasets, and image pre-processing techniques used, development of DNN models, and performance measures used to assess the results.	methodical evaluation process for looking for and choosing research from reputable sources that will guarantee the validity and standard of the chosen material	Publicly availed BrC images dataset is used	to assess how well the generated breast cancer categorization models perform	Ten new research problems and objectives for those who wish to pursue DNN-based BrC image categorization in the future.	2020
5	X. Zhang, Y. Zhang, Q. Zhang, Y. Ren, T. Qiu, J. Ma, Q.	To reflect patients' whole clinical history, this research proposed a detailed framework.	It proved to have greater accuracy than traditional machine learning methods, encouraging its application	Includes 100 breast cancer patients' encounter notes, surgery data, histology notes, imaging.	The lenient micro F1 score is the main assessment criteria. There in dataset, we utilized 10-fold cross-validation to refine the parameters of the model.	Data analysis from health records of just 100 patients might not always fully account for yet more diverse clinical theories and features.	2019
6	Al-antari et al M. A., Han, S. M., & Kim, T. S.	An integrated CAD system of deep learning detection and classification is proposed aiming to improve the diagnostic performance of breast lesions from the entire digital X-ray mammograms.	YOLO detector, Inception ResNet-V2, ResNet-50, and feedforward CNN	DDSM, INbreast	Use of Mammograms	Need to develop a feasible CAD system for practical breast cancer diagnosis.	2020

7	Shen, L., Margolies, L.R., Rothstein, J.H. et al.	Breast cancer detection using mammograms for screening.	Create a deep learning system based on the CNN model.	CBIS-DDSM and INBreast	In the S10 test set, confusion matrix studies were performed on the Resnet50 and VGG16 patch classifiers.	Worked on ROI Annotated factor tuning; it is not adjusting other image patches and does not detect absent ROI basis pictures.	2019
8	Vakanski, A., Xian, M., & Freer, P. E.	Including visual salience in a comprehensive research model for breast tumor segmentation	Create a new deep learning algorithm that utilizes attention blocks and the U-Net architecture.	Breast Ultrasound Images	Dice similarity coefficient (DSC) is used for evaluation of model	Selected attention blocks neglect others, increasing the likelihood of information loss.	2020
9	Khan, M. A., Ashraf, I., Alhaisoni, M., Damaševičius, R., Scherer, R., Rehman, A., & Bukhari, S. A. C.	Deep learning will be used to automate multimodal tumor categorization.	Create a new deep learning system with edge-based histogram equalization and discrete cosine transformations,	BraTS	The robust covariant features based on partial least squares (PLS) were combined into a single matrix	Extreme learning was used, which made things more complicated while ignoring classification variation.	2020
10	Sahlol AT, Abd Elaziz M, Tariq Jamal A, Damaševičius R, Farouk Hassan O.	To extract useful characteristics from photos to identify TB in the chest.	Create a new hybrid technique that combines Mobile Net, a CNN model, and Artificial Ecosystem-based optimization.	Shenzhen and Dataset 2	Fitness measurements include accuracy, sensitivity, specificity, and time consumption.	Used feature selection, leading to the loss of data, and feature selection does not rely on convolution. Decrease the nonlinearity mapping effectiveness of features as a result.	2020

Chapter 3

Material and Methods

3.1 Dataset

The dataset for the study is collected from MIAS image database which we acquired from Kaggle website. The database contains 322 mammograms in 161 set of pairs which have both left and right mammograms of patients. The mammograms are digitizing 50-micron pixels which are layer reduce to 200-micron pixels. To get 1024 x 1024 pixels image in Portable Gray Map (PGM) file format [33]. As there are some missing values they are removed from the dataset. The database is divided into different categories according to specific characteristics. According to type of background tissue it is categorized into three types shown in table 3.1.

Table 3.1 Character of background tissue

Category	No. of Mammograms
Fatty	106
Fatty glandular	104
Dense glandular	112
Total	322

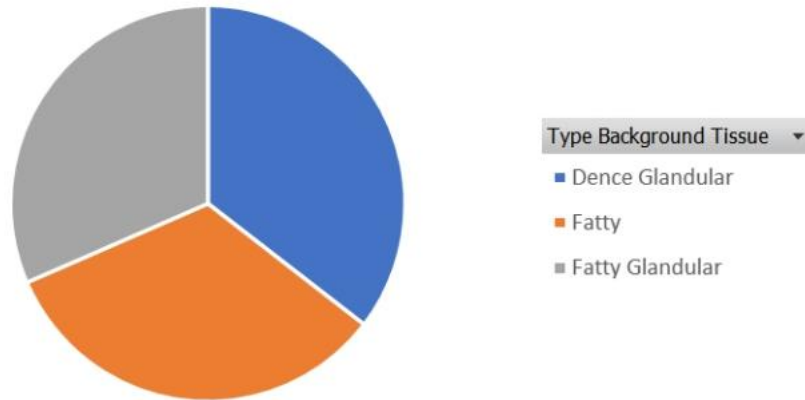


Figure 3.1: Pie Chart description of type of tissues

According to type of abnormality mammograms are divided into seven categories which are as follow:

1. CIRC - Well-defined/circumscribed masses (asymmetrical lumps on the area of the breast)
2. SPIC - Speculated masses (abnormalities with not well-defined lumps)
3. NORM - Normal (displays a typical picture with no anomalies)
4. MISC - Other, ill-defined masses (a variety of anomalies)
5. ARCH - Architectural distortion (Breast tissue architectural deformities are caused by unusual tissue organization)
6. CALC - Calcification (calcium carbonate accumulation around tissues)
7. ASYM - Asymmetry (the mass density of breast tissue is increased)

The number of mammograms that fall under the above categories are represented in table 3.2 below.

Table 3.2: Class of abnormality present

Category	No. of Mammograms
CIRC	24
SPIC	24
NORM	189
MISC	18
ARCH	12
CALC	34
ASYM	21
Total	322

Based on the results of mammography scans, these classifications are further subdivided into the degree of the abnormality present among the tissues. There are two levels of severity.

- Malignant
- Benign

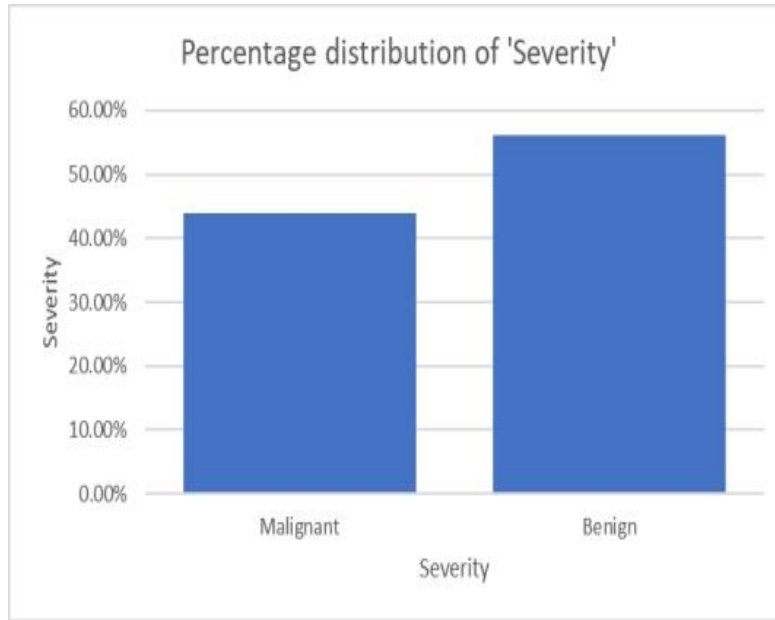
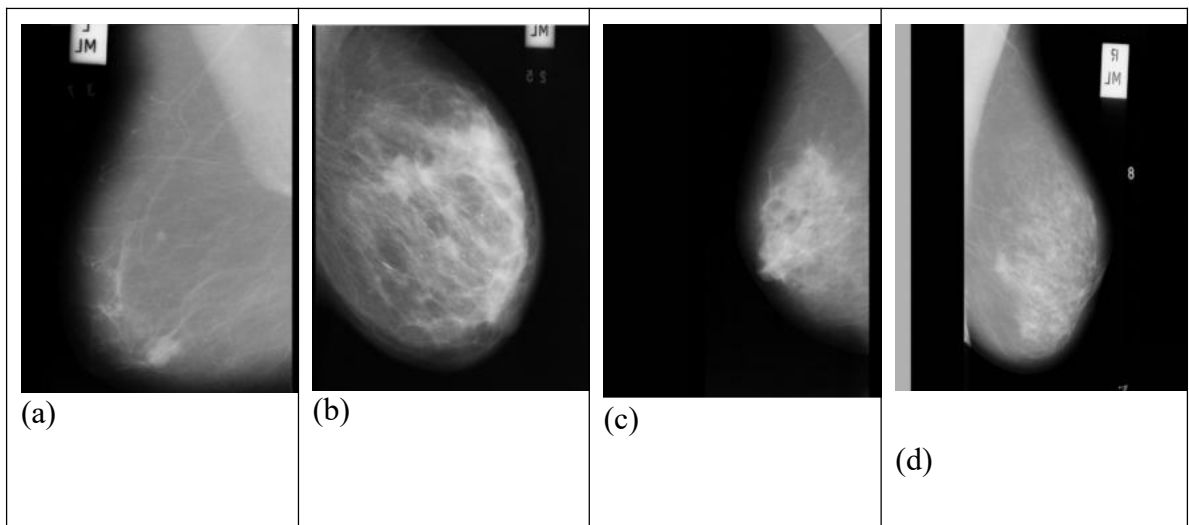


Figure 3.2: Bar chart for severity of cases

There are 54 Malignant, 67 benign and 201 normal mammograms scan present in the dataset. Figure 3.3 shows some examples of mammograms.



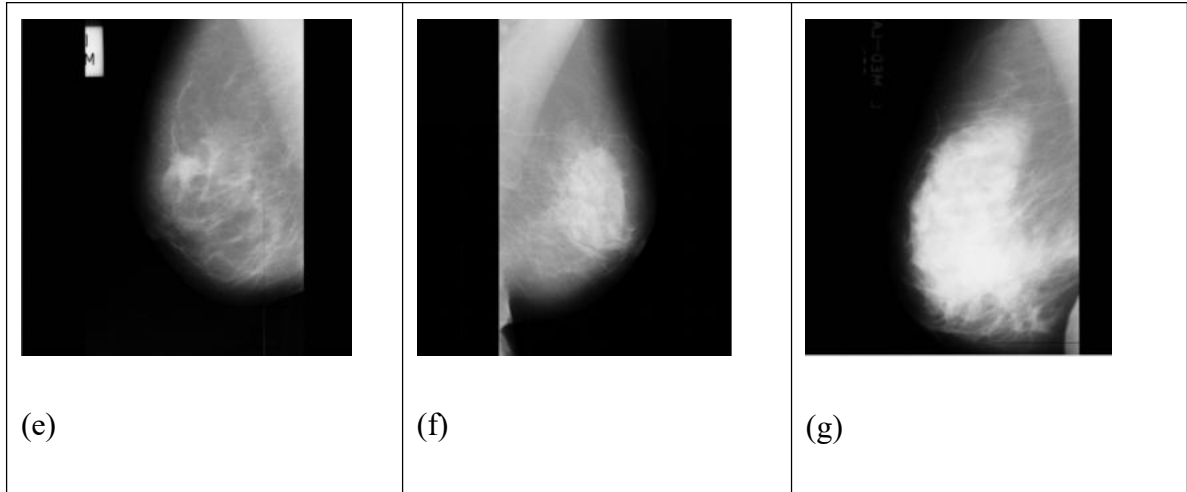


Figure 3.3: Sample images from the mini MIAS database. (a)CIRC (circumscribed masses), (b) SPIC (Spiculated masses), (c)NORM (normal image devoid of masses), (d)MISC (miscellaneous/ill-defined masses), (e)ARCH (architectural distortion), and (f)CALC (calcification), (g)ASYM (asymmetry)

3.2 Training and Testing

The dataset is then separated into two groups for training and testing purposes training data and validation data. 75% of the shots in each class are used to train the network, with the remaining 25% used to validate the trained network. This results in 90 photos for validation and 232 images for training. There are 133 photos of instances with anomalies and 189 pictures of normal cases among the 322 pictures. Classes are classified according to the anomaly present. This dataset partitioning is completely random to achieve impartial network training. The classes are additionally balanced using randomization to accomplish impartial classification for both systems during training.

3.3 Methods Used

The methodology includes preprocessing, lesion identification, feature extraction, and classification. The suggested system accepts a mammography picture as input. It begins by preprocessing the picture by decreasing noise, eliminating artifacts, increasing contrast, and stamping down the pectoral muscle region from.

The methodology is shown in figure 8 below.

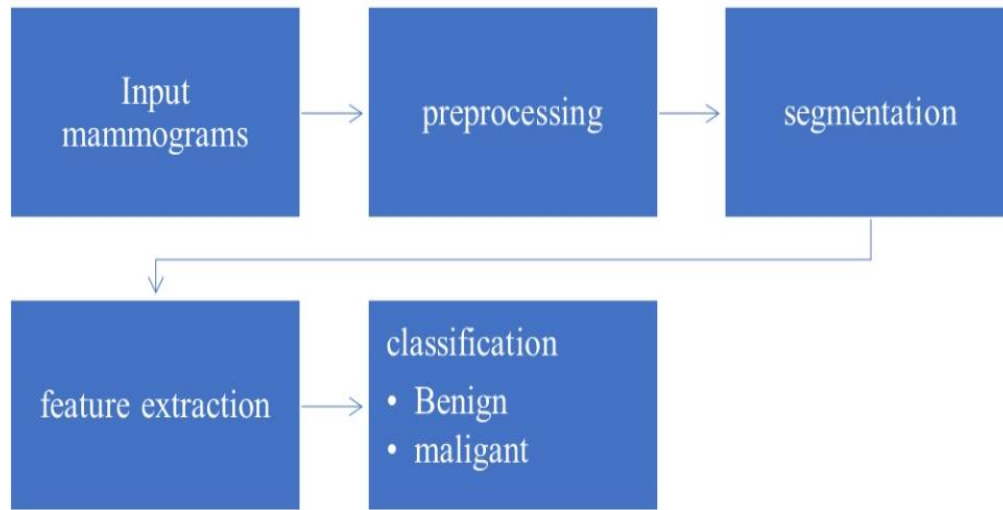


Figure 3.4: Flow diagram of methodology.

3.4 Preprocessing

Preprocessing is a key stage in a mammography classification Modeling system. It aids in the preparation of the mammography picture for subsequent examination and in narrowing the hunt for anomalies. To evaluate the relevance of the preprocessing stage and state that when a preprocessing step is used, the best lesions detection performance is obtained [34]. The preprocessing procedure seeks to eliminate extraneous data from the mammogram picture and improve image quality. Several algorithms and filters are combined in the suggested preprocessing technique. It is divided into four stages: 1. Noise reduction 2. Enhancement of contrast 3. Suppression of labels 4 Removal of the pectoral muscle.

The removal of the pectoral muscle is an efficient method of identifying breast cancer. In the mammography, the pectoral muscles look darker than the rest of the breast tissues. As a result, it is required to find and remove the pectoral

muscle [35], which appears with the same intensity as most anomalies and should thus be eliminated for proper classification.

3.5 Segmentation

In segmentation preprocessed image data is separated into various sections known as segments. This lead to the region of interest (ROI), commonly referred to breast area, being separated from the background and the interior breast tissue features. ROIs are quite helpful since it is preferred to focus on the areas at which real masses might have been found instead of processing the full image. Pretrained models ResNet and VGGNet are used for this purpose. The most popular method is to substitute the network's layers that are completely linked at the end with convolutional layers that output segmentation maps rather than likelihoods of classes. This enables the algorithm to make pixel-by-pixel predictions [36].

3.6 Feature Extraction

CNNs are often used to extract features from photos. Deep CNN architectures like VGGNet and ResNet are shown to be excellent in learning hierarchical features from pictures. These networks are made up of many layers of convolution and pooling that extract features at various degrees of complexity [37]. By processing the mammograms through these already trained models, you may make use of the learnt graphical representations of features without having to train extensively on your unique dataset. You may fine-tune the network's parameters to better capture the distinguishing aspects in your photographs by fine-tuning.

3.7 Data Augmentation

To allow deep learning models, illustrations were enhanced through data augmentation [38]. Deep learning models produce promising outcomes when

evaluated on massive amounts of data. Three mathematical operations are implemented in this work: left to right flip, up to down flip, and rotation at 90 degrees shown in figure 3.5. Table 3.3 shows the number of samples used for augmentation.

Table 3.3: Data Augmentation

Dataset	Training sample	Testing sample	Total	No of images after augmentation process
MIAS	257	65	322	1028

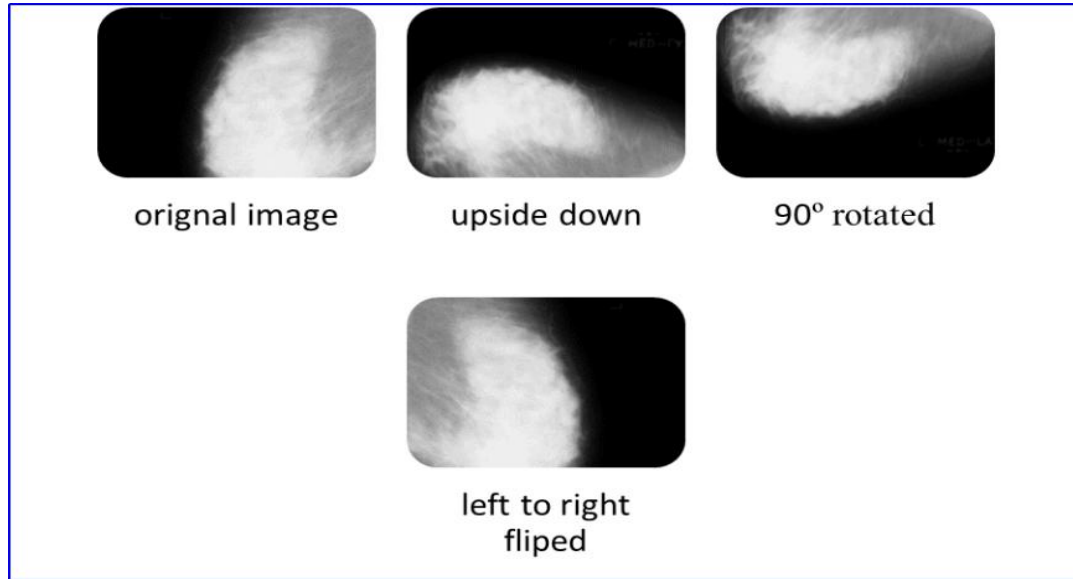


Figure 3.5: Augmented Images of Mais dataset

3.8 Classification

Following the extraction of characteristics from mammography ROIs, these features are used as input or supplied to the classifier to categories the ROI as benign or malignant. This method is known as data mapping to preset classes. Both CNNs classifier is utilized in this work because it can manage high-dimensional feature

vectors. The back-propagation CNN contains one input layer, two hidden layers (each with ten nodes), and one output layer [39]. Every already trained network is partly trained by training certain layers and freezes others. While the lowest layers of each pre-trained network's parameters are left unaltered since these levels have collected general information from the pre-training procedure, while updating the higher layers of the model, which focus more on a particular classification job. As a result, we may fix (i.e., freeze) the weights of layers while fine-tuning (i.e., retraining) the remainder of the layers using the mammography dataset to fit our classification goal [40].

3.9 Evaluation matrix

Accuracy, sensitivity, and specificity were used to assess classification performance on test images. The metrics were calculated for each class, and the mean value of the two classes was used to evaluate the classifier's performance [41]. Accuracy, sensitivity, and specificity may be computed using the following formulas:

$$\text{Classification Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Sensitivity (recall)} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{F1 Score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (5)$$

Here TP stands for true positive, TN stands for true negative, FP stands for false

positive, and FN stands for false negative. The correct prediction of classes is given by TP and TN, whereas the incorrect prediction is given by FP and FN.

The accuracy of deep learning algorithms is frequently used to evaluate their outputs since it displays the entire association. This measure, however, seems unlikely to be enough on its own. This necessitates the development of an alternate measure that is spared by the issue of imbalanced datasets. In the outcome, the Matthews correlation coefficient (MCC), a contingency matrix approach for obtaining the Pearson product-moment correlation coefficient, which is the correlation from actual and projected scores, also served to evaluate the effectiveness of the binary approach for classifications.

The MCC value may be determined as follows:

$$MCC = \frac{TP \times TN - FP - FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \quad (6)$$

While an MCC equation was used with the parameters in the confusion matrix the least favorable score is -1 and the most significant score was +1.

Table 3.4: Evaluation result of CNN model

Classification Accuracy	F1 score	precision	sensitivity	Specificity	MCC value	Training time (m)
96%	96 %	95%	97%	95%	91%	12

Table 3.5: Evaluation results of ResNET50

Classification Accuracy	F1 score	precision	sensitivity	Specificity	MCC value	Training time (m)
97%	97 %	98%	97%	98%	95%	25

Table 3.6: Evaluation result of VGGNET 16

Classification Accuracy	F1 score	precision	sensitivity	Specificity	MCC value	Training time (m)
98%	98%	99%	98%	99%	97%	10.25

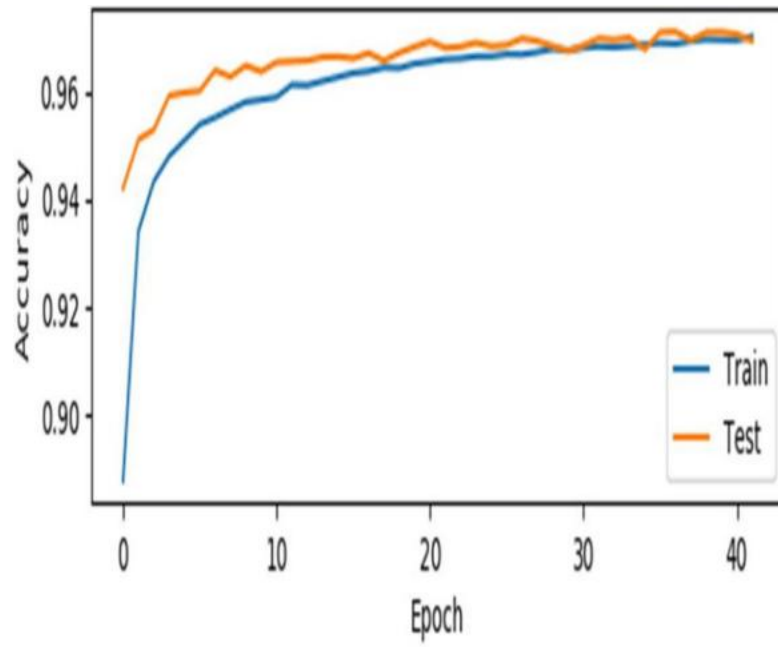


Figure 3.6: Accuracy plot of ResNet50

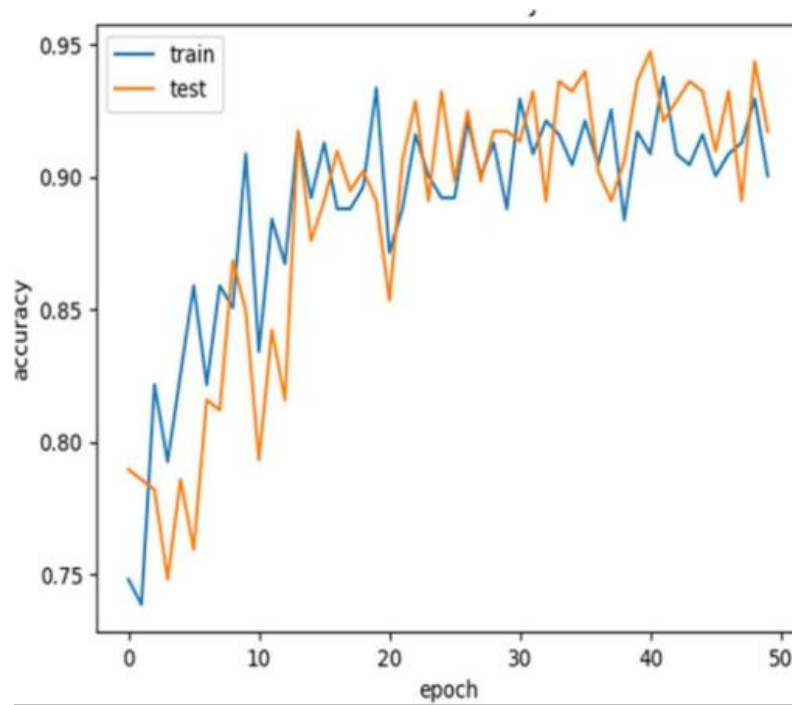


Figure 3.7: Accuracy plot of VGGNet 16

According to Tables 3.4, 3.5, and 3.6, VGG-16 outperformed the other three kinds of CNNs in terms of precision for classification and MCC. A heat map, also known as a type of activating chart, was later produced using Grad CAM. after determining the optimal method, which was VGG-16.

To make the images more apparent and understandable, the recommended VGG-16 training images were incorporated within the Grad-CAM algorithm, which focuses on the way the AI system recognizes, i.e., the way photos are categorized after being programmed with neural network techniques. Grad-CAM takes advantage of the feature maps generated by a CNN algorithm's last convolution layer. Advanced characterization and precise mapping information ought to be clarified most effectively by the final layer of convolution. Grad-CAM visualized the VGG16 algorithm's areas of interest (ROIs) in the test photos, which were correctly defined with an accuracy of 98%.

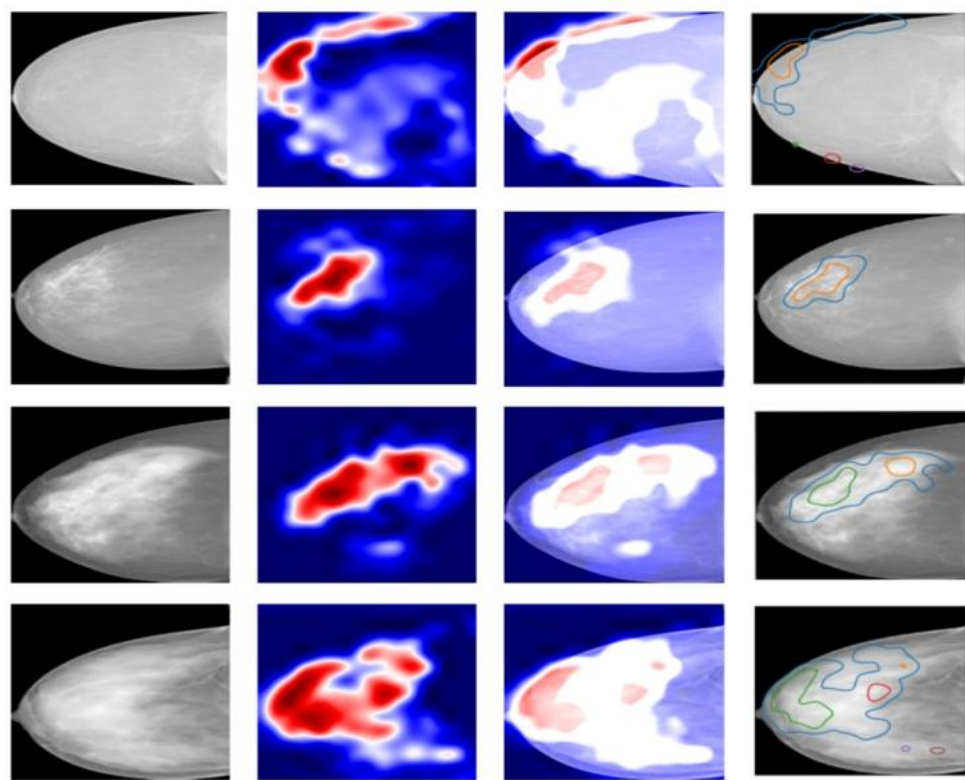


Figure 3.8: Heat generated by GradCAM.

3.10 Explainable Artificial Intelligence survey

In the survey we conducted, we used two parameters: the XAI Explanation Satisfaction scale and the Cahour-Forzy (CHF) scale. The initial study explicitly probes radiologists' impressions of the XAI system's predictability, dependability, and efficiency [42]. The last section specifically asks respondents if they believe the XAI system to be trustworthy, foreseeable, effective, and believable [43]. The opinions regarding each question taken from the CHF scale and the XAI Explanation Satisfaction scale are shown separately in Tables 5 and 6. Radiologists who professionals in their areas of expertise are addressed the questions that follow.

Table 3.7: Statistics of the professional survey's rating using the CHF scale.

No.	Survey Question	Average Response
Q1	How confident are you in the artificial intelligence system?	3.8
Q2	Are the AI technique's behaviors predictable?	4.2
Q3	Is the artificial intelligence system dependable? Do you think it is safe?	3.9
Q4	Do you believe the artificial intelligence system is efficient in its operations?	5.3

Table 3.8: result of XAI satisfaction Scale.

No.	Survey Question	Average Response
Q5	I have trust in the artificial intelligence system. It is functioning properly.	3.5
Q6	The artificial intelligence system's results are quite predictable.	3.6
Q7	The artificial intelligence system is extremely dependable. I can always rely on it to be correct.	2.9
Q8	When I put my confidence in the artificial intelligence system, I feel certain that I will obtain the correct answers.	4
Q9	Because it operates quickly, the artificial intelligence algorithm is efficient.	3.9
Q10	I am frightened about artificial intelligence.	4.7
Q11	Artificial intelligence outperforms a beginner human user in medical image analysis.	4.5
Q12	I like to make judgments using an artificial intelligence system.	3.8
Q13	Artificial intelligence considers all relevant facts to assist me in deciding.	2.7

Q14	Artificial intelligence enables me to make quicker judgments.	3.4
Q15	XAI assists me in making decisions.	3.7
Q16	The system's overall detection and uptime performance is excellent.	4.3
Q17	This approach may be included into a larger healthcare support system, such as the Internet of Things (IoT).	4.1
Q18	This technology has the potential to be employed in further therapy and diagnostic procedures.	2.9
Q19	This model is ineffective in detecting cancer disease.	3.2
Q20	Do you trust the XAI System?	4.9

Table 3.9 Evaluation Summary of Expert Survey

Age	Experience in years	Artificial intelligence experience		Prediction	
35.6	9.7	yes	no	true	false
		20%	80%	30%	70%

Before the questionnaire was administered, expert radiologists were questioned their age, years of experience in the field, if they had previously done any work on artificial intelligence, and whether they were now engaging in such a project. Table 10 shows that the average age was 35.6, the average year of professional experience was 9.7, and just 20 percent had any expertise with artificial intelligence.

Following that, four graphical displays of breast cancer prediction by selected models within the scope of the study were shown, and radiologists were asked to predict the illness. In the survey research, every radiologist recorded the illness. In the end, just 30% of radiologists accurately recognized the illness, while 70% guessed it wrong or left it blank.

Chapter 4

Results and Discussion

4.1 Results and Discussion

The two scales' reliability and normal distribution were determined. The mean of the CHF scale is 4.30 and the XAI Explanation Satisfaction scale is 3.76, as reported in Table 4.1. This suggests that Expert Radiologist will find the Grad-CAM application's outputs sufficient and valuable in the future.

Table 4.1: Normal Distribution and reliability scores.

Scale	mean	Standard deviation	median	Cronbach's alpha (α)	skewness	Kurtosis
CHF	4.30	0.688	4.05	1.177	0.6582	-0.690
XAI	3.76	0.651	3.75	0.514	0.0814	-0.742

Skewness and kurtosis tests were used to determine whether a distribution is normal. It is worth noting that skewness reflects the degree of distortion in a distribution, whereas kurtosis reveals how many or how few samples occur in the center of the distribution. Our dataset corresponds to the normal distribution, as indicated by the values for kurtosis and skewness in Table 4.1, which ranged within the values of -1 and 1.

Cronbach's alpha (α), which assesses item internal cohesion, was used to confirm the consistency of the items in each scale. A high Cronbach's alpha indicates that the items on a scale are consistent with one another and measure the same trait. According to the findings, the CHF scale (=1.177) and the XAI Explanation Satisfaction measure (=0.514)

were also satisfactory. Furthermore, a considerable positive correlation was discovered between the two scales.

Table 4.2: Regression Model Summary.

(n=100) XAI						
Survey	St. Error	B	ρ	R^2	Adj R^2	F
CHF	0.15	0.85	0.1	0.53	0.50	9.1

The Summary regression outcome is shown in Table 4.2. The independent parameter of the CHF scale ($=0.85$, $S.E.=0.15$) has a substantial positive influence on the dependent parameter of the XAI Explanation Satisfaction scale. The independent factor of the CHF scale explains 50% of the variation in the XAI Explanation Satisfaction scale (Adjusted $R^2 = 0.50$).

Table 4.3 Difference table.

Less than = 4			Grater then =5			
Survey	St. Div	Mean	St. Div	Mean	p	T
XAI	0.48	3.74	0.48	4.84	0.16	-1.69
CHF	0.48	3.87	0.87	6.18	0.2	-3.13

As demonstrated in Table 4.3, there was indeed a significant difference in the mean of the CHF scale in terms of professional experience ($t=-3.13$, $p0.05$). As a result, the mean of

the Respondents with a total of five years or greater of work expertise ($X=6.18$, $S.D.=0.87$) had a greater mean ($X=4.84$, $S.D.=0.48$) than those with four years of expertise or fewer ($X=3.87$, $S.D.=0.87$). There was not any significant change in the mean of the parameter of the XAI Explanation Satisfaction scale ($p>0.05$).

In terms of application, having found that the CNN models scored at least 95% on the three criteria, particularly against the novel and original dataset on breast cancer disease, is amazing. However, this conclusion does not immediately convey the system's level of safety and comprehension. The findings in the assessments were then interpreted in the context of artificial intelligence and human-machine interaction, which may be explained by several distinct radiologists.

Grad-CAM was subsequently implemented to the outcomes of the VGG16 model, which has the highest the present research and appropriate in the framework of the open topic). The establish classification accuracy. The radiologist assessed the heat maps developed as extremely accurate. Positive comments were made on performance and XAI. This technique may be discussed without regard to the detection performance of breast cancer. In other words, the XAI system has been proven to be a successful instrument in the field of health due to its human safety and efficacy.

Furthermore, within the framework of open issue applications, various parametric testing (including hyperparameter optimization) might be assessed on the VGG16 model. Whilst the images chosen for the research are recommended for an equitable approach application may be developed using additional databases made up of all images from myriad studies or a similar number of distinct images. Apart from the selected pictures, alternative pictures can be employed (this is both that lie within the realm of possibility of heed CNN-VGG16/Grad-CAM system is suitable for different medical scans in addition to mammographic images.

Modern Technologist today are also progressing towards the Internet of Things (IoT) and, in this regard, the Internet of Health Things (IoHT) [44]. Because of the explainability

aspect that users want, this approach creates a GUI for an everyday IoHT system and creates a productive hardware-software combination.

4.2 Requirements for Explainable Artificial Intelligence System development.

The European Union has identified seven important prerequisites for AI models to be trusted [45].

4.2.1 Obligations and Conduct

The healthcare sector has increased its demands for the implementation of AI, and determining how to define the principles and responsibilities of AI has become a concern. For example, reckless AI might result in the loss of medical professionals and patient records [46]. In addition, it entails ethical concerns about confidentiality of information when using AI models. Now, AI scrutiny and transparency are in their early phases. Furthermore, the idea of responsible AI is examined and analyzed to build accountability concepts for technical domains [47]. As a result, explainability may be a necessary requirement for defining certain AI roles.

4.2.2 Security and Secrecy

AI systems demand real-time data updates, which poses a systemic danger. AI's black-box nature may lead to a slew of security issues, which can arise from both internal and external sources. These issues might be internal to the algorithm or external, such as poor user usage and the fabrication of bogus datasets by network assaults [48]. The authors of [49] identified three categories of black-box artificial intelligence security risks: network assaults, system bias, and mismatch attacks. These risks have the potential to have major effects for medical systems. For these factors, various research has explored and offered security of data strategies for XAI models [50].

4.2.3 Transparency and Skepticisms

An AI model is developed using datasets with unique intrinsic characteristics, which include concealed bias. In age and skins-color identification algorithms utilizing gathered

photographs of people of diverse ages and races, for instance, the AI system preferred white skin and a 45-year-old individual [51]. Additionally, unavoidable hazards in large data techniques allow for greater bias and duplication, as well as worsening human mistakes [52]. Particularly, the bias might be caused by data, algorithms, or responses from users [53]. This will have a significant impact on the fairness of AI models, resulting in varied outcomes for different people. In medical practice, various patients may exhibit distinct symptoms that influence the methodology.

4.3 Pros and Cons of Explainable Artificial Intelligence

While systems based on XAI provide considerable benefits in terms of openness, confidence, and decision-making support, it also presents obstacles in terms of complexities, analysis: trade-offs, and user comprehension. The creation and use of XAI systems necessitate a careful evaluation of these benefits and drawbacks, as well as continuing research and developments in XAI methodologies to meet the limits [54].

Table 4.4: Pros and Cons of XAI

Pros	
Disclosure	When contrasted with black-box artificial intelligence models, XAI systems give interpretations for their judgments and predictions, thereby rendering them more readily apparent. This openness allows consumers to figure out why a particular choice was taken, increasing confidence, responsibility, and adoption of AI systems.
Credibility And Reliance	The interpretability given by XAI algorithms inspires certainty and credibility in end-users such as medical personnel, government agencies, and those impacted by AI choices. Recognizing the logic behind AI forecasts can lead to increased acceptance and adoption

	of AI technology.
Improved Decision Making	XAI models provide an understanding of the characteristics, trends, and factors influencing AI judgments. This knowledge can help with processes for making decisions, enabling users to make better educated and assured judgments upon reviewing the system's explanations.
Testing And debugging	By allowing users to analyze their decision-making process, XAI platforms may assist in verifying the efficacy and dependability of AI systems. If a machine learning algorithm generates unanticipated or inaccurate results, the explanations supplied by XAI might help discover and troubleshoot any model flaws or biases.
Moral And Regulatory Compliance	XAI algorithms can help ensure conformity with ethics and law by allowing users to examine and verify if AI judgements are in accordance with accepted standards, legislation, and equitable norms. This is especially crucial in fields like medical care, financial services, and law enforcement.
Cons	
Complex And Expense	Using XAI methods can add difficulty and computational costs to AI systems. Creating insights and retaining model interpretability may necessitate more computing time and resources, which can have an influence on system speed and capacity.
Difficulties With Interpretation:	In complicated algorithms with data that is highly dimensional, XAI approaches may occasionally give easy or logical answers. To completely grasp and interpret the interpretations provided by XAI

	systems, experts and subject knowledge may be required.
Productivity Vs. Interpretability Trade-Off	There may be a trade-off between AI model interpretability and productivity. When contrasted with more complicated and less interpretable models, highly interpretable models may forfeit some amount of performance. A fundamental problem in designing XAI systems is striking a balance between interpretability and efficiency.
Narrow Scope of Description	Artificial intelligence predictions may not always be explained completely or comprehensively by XAI algorithms. Because of the intricate design of deep learning models and the characteristics of some algorithms, providing full explanations for every choice may be difficult, resulting in incomplete or condensed interpretations.
User Comprehension and Bias	The efficiency of XAI technology is dependent on the capacity of users to comprehend and grasp the reasoning supplied. Users' degrees of knowledge may vary, and they may understand information a unique way, thereby injecting biases or misinterpretations into decision-making.

Chapter 5

Conclusion

5.1 Conclusion

Nowadays, artificial intelligence algorithms are widely and successfully applied in a variety of fields, such as chatbots, self-driving cars, and facial recognition. However, there are significant concerns regarding how current algorithms make decisions, or the "black box" issue. There are substantial problems with the explanation and interpretation of these designs, even if the mathematical equations and concepts for AI-based algorithms are solved. For instance, decision trees provide a prominent level of explanation. However, they fall short of deep learning algorithms in terms of classification accuracy.

The VGG16 model, a particularly precise classifier among the three CNNs, is heat-mapped in this study using Grad CAM. Based on this, clinicians were given a serious study idea concerning where machines truly focus during categorization and if these focal locations are accurate. Based on the findings of a survey of radiologists, the artificial intelligence system concentrated on the right locations, specifically, the same persons throughout categorization.

Consequently, the algorithm produced a highly efficient result. This indicates the VGG16 model's excellent efficiency. It is reasonable to claim that we supplied a competent and explainable diagnostic tool for the identification of breast cancer using the established system heat maps.

5.2 Recommendations

In the future, it will be critical to promote seamless and successful interaction between humans and machines in artificial intelligence-based networks, particularly in healthcare diagnostic systems.

Furthermore, a transparent interaction built on trust is required. Future systems built on artificial intelligence ought to mirror the way users develop and transmit explanations to every event faced in their daily lives while giving the essential explanation to users.

This research establishes the groundwork for future research on alternative diagnostic methods, and it significantly supplements the academic literature with diverse assessments based on explainable artificial intelligence approaches.

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